

Can Language Models Simulate Psychometric Respondents?

A Cross-Vendor Sensitivity Study with Baseline Controls and a Replicated Zero-Prompt Framing Manipulation for a Polish Mentalization Battery

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Abstract

Before collecting human data for a new psychometric battery, I test whether the instruments *react* to personality variation at all. Thirty fictional Polish-language biographies with predetermined psychological profiles were processed by five large language models from four vendors — Claude Sonnet 4.6 and Opus 4.6 (Anthropic), GPT-5.4-mini (OpenAI), Grok-4-20-reasoning (xAI), and Gemini 3 Flash (Google). Each persona was attempted twice per model (300 attempts; 285 valid persona runs after exclusion of partial outputs and content-filtered responses), and 7 human respondents completed the same battery (292 observations including human pilot). This is a **sensitivity study**, not a validation.

Four published self-report instruments showed consistent profile recovery: attachment $\kappa = .33-.87$ across models; rank-score correlations median $r = .71-.82$; literature consistency 8–10/10. Cross-vendor questionnaire agreement was high (median $r = .78-.96$), but TCTM-22 cross-vendor agreement was low (ICC = .25). Error profiles were model-dependent: Claude Sonnet produced almost exclusively DOS (97%); Gemini produced the descriptively closest-to-human-pilot error profile.

Two baseline experiments decompose the observed variance into three components. In the *baseline condition* ($N = 46$ valid runs, 7–10 per model), models completed the battery with an “answer as yourself” instruction, revealing stable “default response profiles”: GPT is anxious-preoccupied in 100% of runs, Gemini hits near-maximum on all self-report scales, and different models from the same vendor (Sonnet vs. Opus) differ in attachment style. The *zero-prompt condition* was extended to a replication of $N = 5-8$ runs per model: 31 valid TCTM-22 payloads and 30 parseable self-report payloads (one GPT zero-prompt output had a complete TCTM-22 array but no parseable self-report sections). With multiple observations the “answer as yourself” instruction provides replicated evidence of systematic shifts in self-report profiles: Sonnet’s avoidance dropped from baseline $M = 3.88$ to zero-prompt $M = 2.98$ ($N = 6$, $\Delta = -0.90$, closely replicating the original single-shot estimate of -0.99); Gemini’s emotional stability dropped from baseline $M = 7.0$ to zero-prompt $M = 6.0$ ($N = 8$, $\Delta = -1.00$, exact replication); and four of five models reported *higher* attachment anxiety without the instruction (Sonnet $+1.02$, Opus $+0.55$, Grok $+0.75$, Gemini $+0.90$). Author-keyed TCTM-22 accuracy was preserved across framing conditions for four of five models; GPT dropped from baseline 78.6% to zero-prompt 70.5% (-8.1 pp), the only model showing a notable accuracy shift. Persona manipulation increased response variance $2-51\times$ over baseline stochasticity (model-level medians $3.5-8.9\times$ across 9 questionnaire dimensions where baseline $SD > 0$), confirming that persona-driven variance is not an artifact of inherent model noise.

Core materials (biographies, scored responses, analysis scripts) are released as an open dataset.

Keywords: synthetic respondents, instrument sensitivity, mentalization, attachment, cross-vendor LLM comparison, baseline control, LLM default response profile

1 Introduction

Developing new psychometric instruments requires extensive piloting. A growing line of research suggests that large language models (LLMs) can serve as *simulated respondents*, providing rapid probes of instrument behavior before human data collection [1, 3]. The core question is not whether LLMs “have” personality, but whether instruments *react systematically* to known personality variation embedded in biographical prompts. If they do, the instruments are at minimum sensitive to the relevant signal — a necessary (though far from sufficient) precondition for construct validity [2]. If they do not, there is a problem worth catching before human recruitment.

This framing is important: the present study is a **sensitivity test**, not a validation study in the classical sense. I ask: “Given 30 biographies designed to span the full range of attachment styles, mentalization levels, and personality traits, can the instruments recover the embedded profiles from LLM responses?” A positive answer means the instruments are responsive; it does not prove they measure the intended constructs in human respondents.

Most existing studies use a single model family [1, 3]; by contrast, the present study employs **five models from four vendors** (Anthropic, OpenAI, xAI, Google), enabling assessment of cross-vendor consistency — whether sensitivity findings generalize across architectures or are model-specific artifacts. Crucially, two baseline experiments are included: a *baseline condition* (models complete the battery “as themselves,” repeated 7–10 times each) and a *zero-prompt condition* (models receive the battery with no system instruction at all). This design allows decomposition of the observed persona-driven variance into three components: inherent model stochasticity, the effect of instructional framing, and the genuine persona manipulation effect.

I apply this approach to four published Polish-language instruments — DBZ-R (attachment), MentS-PL (mentalization), KPP (need for cognition), TIPI-PL (Big Five) — and one original mentalization vignette test (TCTM-22). Thirty fictional personas were constructed, processed through all five models with test–retest, and scored against ground-truth profiles. Seven human respondents completed the same battery for exploratory comparison.

2 Method

Personas. 30 biographies in Polish (1,554–2,914 file-total words including YAML metadata; 1,489–2,861 narrative-body words after stripping YAML), written in second person, encoding predetermined profiles across 12 dimensions (11 continuous + 1 categorical): attachment anxiety and avoidance (DBZ-R), mentalization self/other/motivation (MentS-PL), need for cognition (KPP), Big Five E/A/C/ES/O (TIPI-PL), and attachment style (4-category classification). Biographies were authored by the first author using Claude Opus 4.6 as a writing assistant; the author specified profiles, narrative constraints, and stylistic requirements, reviewed and edited all drafts. Six biographies (adrian, ewa, kuba, magda, pawel, weronika) were revised after pilot review and are tagged `persona_version: v2` in the

released YAML; the remaining 24 are at v1. The version that was actually sent to all five models is the version referenced throughout this paper and recorded in `all_data.csv`. No biography was designed to contain trait labels; the first author manually coded biographies into two groups based on whether they contained any direct trait-language references (e.g., explicit mentions of attachment style, introversion, openness): nine biographies were flagged as “explicit” (ania, anna-sim, filip, jakub, kamil, marek, natalia, ola, tomek) and 21 as “narrative”. A separate regex-based audit using a broader keyword set yields a different split (see Trait Transparency in Results); both codings are released in the supplementary archive.

Instruments. DBZ-R (Polish adaptation of ECR-R; 36 items, Likert 1–7; Lubiewska et al., 2016), MentS-PL (28 items; Jańczak, 2021), KPP (36 items; Matusz et al., 2011), TIPI-PL (10 items; Sorokowska et al., 2014). These four instruments use published Polish norms. TCTM-22 (22 original vignettes with MASC error taxonomy [4]) is an original instrument without published norms; it is scored against author-defined answer keys. The 22 vignettes cover five subscales (4–5 items each): *Rozpoznawanie podtekstu* (subtext recognition; 4 items: w01, s07, s08, s10), *Rozpoznawanie manipulacji* (manipulation recognition; 4 items: w08, c07, c10, w11), *Wykrywanie sygnałów wygasania więzi* (waning-bond detection; 4 items: w13, w14, e08, w15), *Dynamika dążenie–wycofanie* (pursuit–withdrawal dynamics; 5 items: w19, pw07, w22, pw09, pw11), and *Ocena autentyczności naprawy relacyjnej* (authenticity of relational repair; 5 items: w25, r08, w28, r09, r10).

Ground truth and STYLE_MAP. Each persona’s expected attachment style is taken from the top-level `expected_style` field in `analysis_summary.jsonl`. Four personas (Ewa, Kamil, Marek, Radek) are coded as disorganized in the YAML `expected_profile` block of their biography file but as fearful_avoidant in the top-level `expected_style` field used by the scoring pipeline; this is intentional and corresponds to mapping disorganized attachment to fearful-avoidant under the four-category Bartholomew taxonomy that all five LLMs use in their style outputs. Computing κ under a strict five-category policy (treating disorganized as a separate fifth class that no model can produce) would lower all attachment κ values by ~ 10 percentage points; the supplementary archive includes both calculations.

Models. Five models from four vendors (Table 1). Each persona was processed once per model (run 1), then re-run under identical conditions (run 2, retest). All models used default sampling parameters. Total: 285 valid persona runs (300 attempted; missing retests resulted from API content filters, partial generations, or incomplete TCTM responses — see Table 1 footnote). Plus 5 additional Gemini reruns (run 3) for personas ania, anna-sim, kasia, marek, michal-sim where the first two attempts returned partial or degenerate output; these run-3 records are included in the released open dataset for full transparency but are excluded from all run 1 / run 2 paired analyses, all cross-model first-run statistics, and all main-text tables and figures, to avoid asymmetric over-representation of Gemini. GPT-5.4-mini returned incomplete TCTM responses (20–21 instead of 22 items) in 20% of attempts; affected runs were excluded from per-model TCTM tables (yielding $N = 27$ for run 1, $N = 21$ for run 2). For other instruments, missing items were scored according to the published convention (DBZ-R / MentS-PL / KPP / TIPI-PL: missing-as-NA, with row-wise pairwise deletion in correlation analyses).

Human pilot. Seven respondents (ages 22–60, 4 male, 3 female; 2 psychology students, 5 without psychology background) completed the same battery online. The target population for eventual validation is young adults (ages 18–35) in romantic or close relationships; one respondent (R07, age 60, female) was

Table 1: Models used. Run counts reflect valid completions after exclusion of partial / content-filtered attempts. The 5 additional Gemini run-3 records are partial-fix retries for personas with earlier degenerate outputs and are reported transparently rather than discarded.

Model	Vendor	Platform	Runs (run 1 + run 2)
Claude Sonnet 4.6	Anthropic	AWS Bedrock	30 + 30
Claude Opus 4.6	Anthropic	AWS Bedrock	30 + 25 [‡]
GPT-5.4-mini	OpenAI	Azure OpenAI	27 [†] + 21 [†]
Grok-4-20-reasoning	xAI	Azure AI Foundry	30 + 29 [‡]
Gemini 3 Flash	Google	Gemini API	30 + 28 [‡] (+5 run-3)
Human respondents	—	—	7

[†]Azure content filter or partial TCTM response. [‡]One or more personas failed run-2 due to transient API issues (timeout, refusal, or partial JSON). For Opus run-2 the missing personas are `anna-sim`, `jakub`, `kasia`, `micah-sim`, `piotr`; for Grok run-2 it is `piotr`; for Gemini run-2 it is `ewa`, `sara`. None were retried under the v4 protocol; the missing-data pattern is disclosed here for full transparency.

notably older than this range and was retained for exploratory purposes.

Ethics. The human pilot used informed consent disclosing study purpose, voluntariness (right to withdraw at any point without consequence), GDPR-compliant data handling (encrypted storage, five-year retention, no cross-linking with personal identifiers), and links to mental-health support resources (Polish national crisis-line numbers). All seven respondents consented before completing the battery. The pilot is non-interventional, uses no clinical population, and was treated as exempt from formal Institutional Review Board approval per UKEN Kraków institutional guidelines for pre-pilot psychometric data collection; this exemption status reflects the author’s understanding of those guidelines and is disclosed here for full transparency.

Baseline condition. Each model completed the battery 7–10 times (46 valid runs total; one Gemini run excluded for incomplete TCTM) with the system instruction: “*You are an AI language model. Answer the questions honestly, as you — an AI — would answer if you had to choose. Do not play any character. Answer as yourself.*” (translated from Polish). This condition measures inherent model stochasticity and reveals each model’s “default response profile.”

Zero-prompt condition. Each model completed the battery $N = 5$ –8 times without any system instruction — only the test content was provided as a user message: 31 valid TCTM-22 payloads (Sonnet 6, Opus 5, GPT 6, Grok 6, Gemini 8) and 30 parseable self-report payloads (one GPT zero-prompt output had a complete TCTM-22 array but no parseable self-report sections). The same TCTM-22 instrument used in the persona condition was administered, alongside the four self-report instruments, in a single user message. Without instructional context, models produced widely varying JSON formats (e.g., Sonnet alternated between TCTM22, `czesc_1_winiety`, and `part1_vignettes` as top-level keys across runs); responses were post-processed with a universal scorer that filters to the 22 standard TCTM items and applies the validated `score_dbzr/ments/kpp/tipi` functions used elsewhere in this study. This condition measures the model’s “raw” response to a psychometric task, before any instructional framing.

Metrics. (1) Attachment style match (κ , Cramér’s V), (2) rank–score correlations (Pearson, Spearman), (3) literature consistency (10 construct pairs), (4) test–retest reliability (Pearson r , ICC), (5) inter-model consistency (ICC, Fleiss’ κ), (6) TCTM-22 error profiles (DOS/NAD/BK), (7) per-item cross-vendor agreement (Cochran’s Q).

3 Results

All cross-model tables report first-run data ($N = 30$ per model except where the per-model footnote on Tables 2 and 6 indicates a smaller valid-output count; specifically, GPT-5.4-mini run 1 has $N = 27$ for analyses requiring a complete 22-item TCTM-22 response). Retest data is used for reliability analyses and, where noted, to confirm systematic patterns.

3.1 TCTM-22 Score Distribution

Table 2: TCTM-22 by group (first run).

Group	N	M	SD	Range	DOS%	NAD%	BK%
Human	7	14.3	1.4	12–16	32	36	32
GPT-5.4m	27 [†]	17.0	1.7	13–19	70	6	24
Grok-4-20	30	18.4	4.2	6–22	64	16	21
Opus	30	19.0	2.1	8–20	86	8	7
Sonnet	30	19.4	0.7	17–20	97	0	3
Gemini 3F	30	19.6	2.1	9–21	23	41	37

[†]GPT returned no valid 22-item response for 3 of 30 personas (Paweł, Michał-K, Bartek).

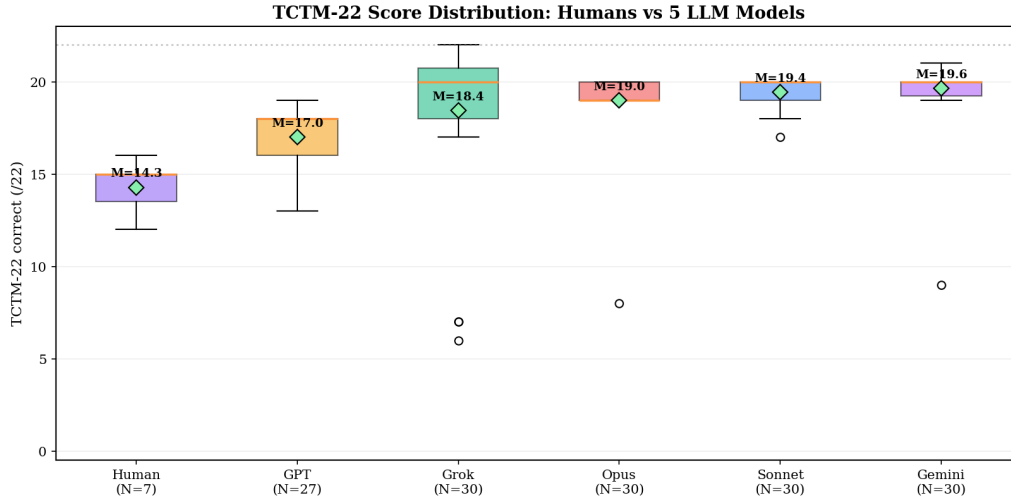


Figure 1: TCTM-22 distribution by group: Human + 5 LLM models. Diamond = mean.

A clear gradient emerged: Human ($M = 14.3$) < GPT (17.0) < Grok (18.4) < Opus (19.0) $\approx$ Sonnet (19.4) $\approx$ Gemini (19.6).

Error profiles were model-dependent. Claude Sonnet produced almost exclusively DOS errors (97%); Gemini produced the closest match to the human pilot profile (DOS 23%, NAD 41%, BK 37% vs. human 32/36/32%). In the human pilot, NAD errors were more frequent than in all LLMs except Gemini, which exceeded the human NAD rate (41% vs. 36%).

3.2 Attachment Classification

3.3 Rank–Score Correlations

Median Pearson r across 11 continuous dimensions (first run): Opus .819, Grok .815, Sonnet .800, Gemini .763, GPT .711. Across all five models, 54/55 dimension-level correlations were positive (expected direction); the single negative case was GPT on MentS-Self ($r = -.06$), where higher expected

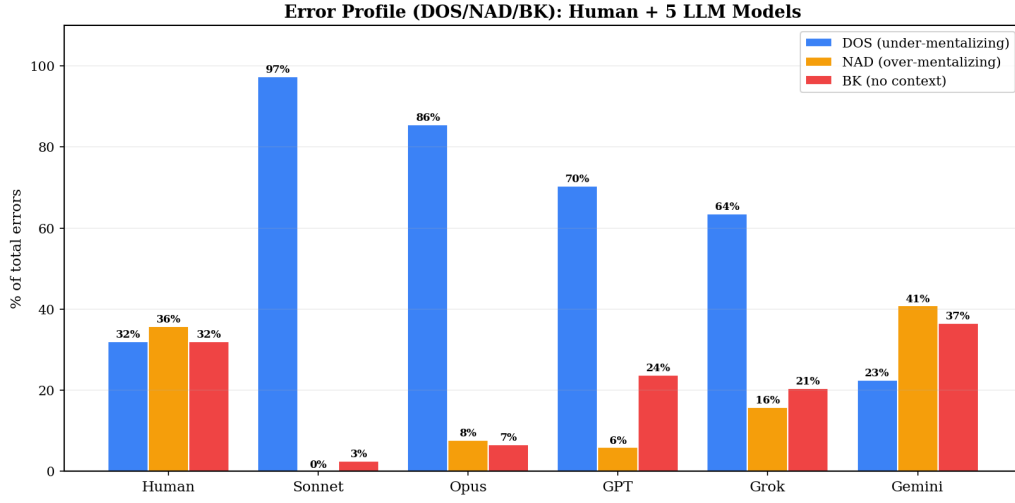


Figure 2: Error type distribution: Human + 5 LLM models. Humans show a roughly balanced profile (DOS 32%, NAD 36%, BK 32%); Claude Sonnet is almost exclusively DOS (97%). Among LLMs, Gemini’s profile (DOS 23%, NAD 41%, BK 37%) is closest to the human distribution.

Table 3: Attachment style classification.

Model	Accuracy	κ	V
Opus	27/30 = 90%	.866	.905
Grok-4-20	27/30 = 90%	.867	.890
Gemini 3F	25/30 = 83%	.778	.801
Sonnet	24/30 = 80%	.735	.765
GPT-5.4m	13/27 [†] = 48%	.333	.474

[†]GPT run-1 with valid 22-item TCTM only ($N = 27$); the 3 personas with missing TCTM (Paweł, Michał-K, Bartek) are excluded from κ computation rather than scored as misclassifications.

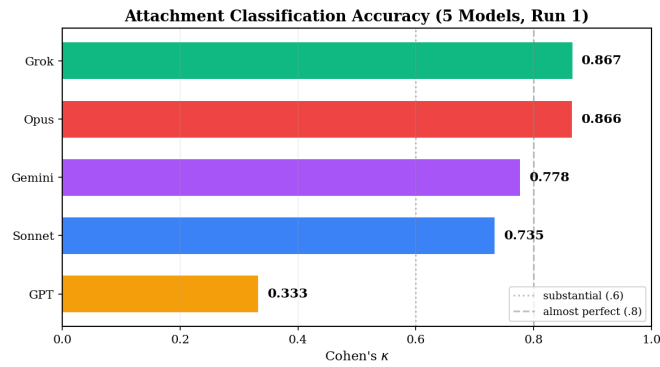


Figure 3: Attachment κ by model (5 LLMs).

mentalization-self failed to track higher self-reported scores. Separately, literature consistency (10 published cross-scale correlation signs): Gemini 10/10, Sonnet 10/10, Opus 9/10, GPT 9/10, Grok 8/10.

3.4 Test-Retest Reliability

Table 4: Test-retest reliability (run 1 vs. run 2).

Metric	Sonnet	Opus	GPT	Grok	Gemini
z-score median r	.981	.993	.890	.950	.962
TCTM total r	.691	.992	.279	.530	.905
Style agreement	29/30	25/25	18/21	29/29	25/28
N paired personas	30	25	21	29	28

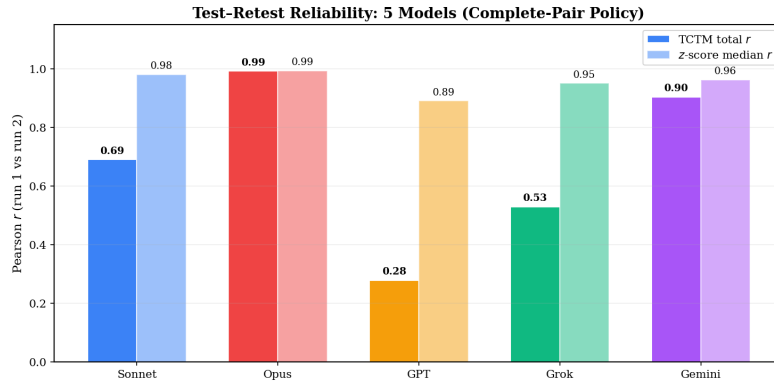


Figure 4: Test-retest reliability across 5 models. Solid bars: TCTM-22 total r . Lighter bars: z-score median r across 9 questionnaire dimensions. Reliability is computed on *intersect-only paired personas* (complete-pair policy, N varies by model).

Opus was near-deterministic across all tasks. Gemini showed high TCTM reliability ($r = .91$). GPT was unreliable on TCTM ($r = .28$ on the 21-persona intersect of valid run-1/run-2 pairs), with individual personas shifting by up to ± 5 points between runs; its questionnaire data was more stable (z-score median $r = .89$). All test-retest correlations are computed under a complete-pair policy: paired correlations use only personas with valid responses in both runs (intersect N varies by model and is shown in the table), rather than imputing missing pairs.

3.5 Cross-Vendor Consistency

Questionnaire z-scores showed high cross-vendor agreement: median inter-model r ranged from .78 (GPT×Gemini) to .96 (Sonnet×Opus). TCTM-22 cross-vendor agreement was low: ICC(2,1) = .254 across 5 models on the $N = 27$ intersect of complete TCTM data. The highest TCTM pair was Gemini×Opus ($r = .94$); the lowest was GPT×Grok ($r = -.05$, effectively zero with a slight negative sign on the intersect).

Fleiss’ κ for attachment style classification (all 5 models as raters, $N = 27$ personas with complete data across all models): .619 (“substantial agreement” under the widely-used categorical-data benchmarks of [6]; we note that those benchmarks have been criticized as arbitrary, and report the raw κ value to permit reader-specific interpretation). Cochran’s Q revealed that 14/22 TCTM items showed significant ($p < .05$, uncorrected, $df = 4$) model disagreement. Under Bonferroni correction at $\alpha/22 = .00227$ (chi-square threshold > 16.64), 8/22 items remain significant: w01, s07, w11, w15, pw07, w22, w28, r10 (all with

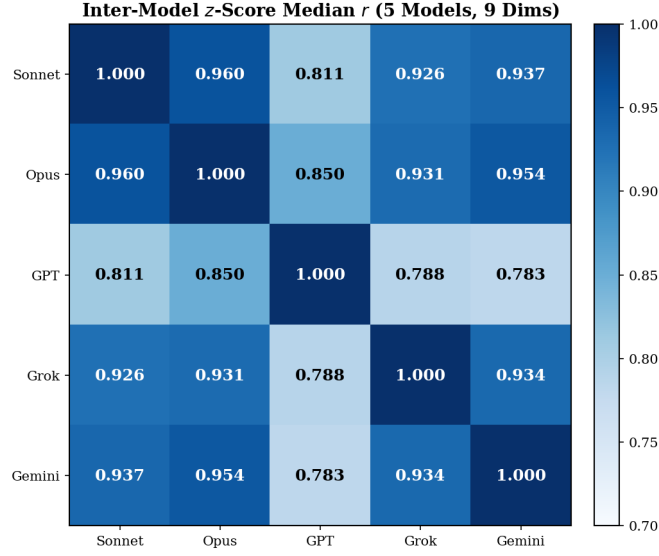


Figure 5: Inter-model z -score median r (9 questionnaire dimensions, 5×5 matrix). Diagonal = 1.

$Q > 32$, $p < 10^{-5}$). Under Benjamini–Hochberg FDR control at $q = .05$, 13/22 items are significant. The full per-item Q , p_{unc} , Bonferroni-pass and BH-pass values are released in the supplementary archive.

3.6 Ola: Cross-Vendor Item Agreement

Persona Ola (“secure but non-mentalizing”) had the lowest cross-model mean TCTM (11.4/22: Sonnet 17, Opus 8, GPT 16, Grok 7, Gemini 9). Cross-vendor error analysis revealed **systematic item-level agreement**: all 5 models produced DOS errors on items s07 and w22. This pattern was confirmed by retest on Opus (8 $\rightarrow$ 9/22) and Gemini (9 $\rightarrow$ 7/22).

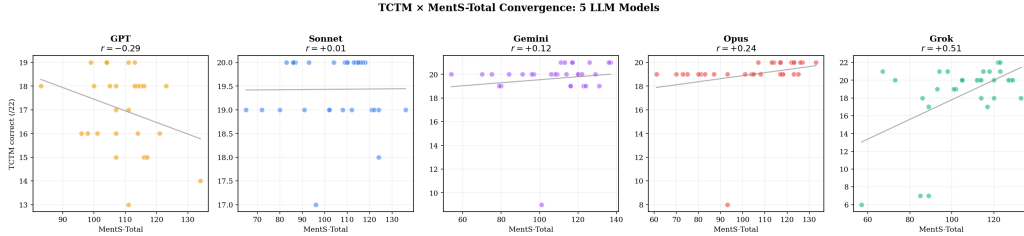


Figure 6: TCTM $\times$ MentS-Total by model. Convergence varies in both magnitude and sign: Grok $r = +.51$ (highest), Opus $r = +.24$ (weakly positive), Gemini $r = +.12$, Sonnet $r = +.01$ (near-zero), GPT $r = -.29$ (weakly *negative*: in GPT, higher self-reported mentalization is associated with lower TCTM accuracy).

3.7 Exploratory Human Comparison ($N = 7$)

Seven human respondents (ages 22–60) scored $M = 14.3/22$ ($SD = 1.4$), well below all LLM means. Human error profiles were distributed across DOS (32%), NAD (36%), and BK (32%). On most author-keyed TCTM-22 items, LLM scores were descriptively higher than the small human pilot: on 7/22 items (w08, c10, w13, e08, w19, w25, w28) the cross-model LLM mean was ≥ 30 percentage points above human accuracy (range +34 to +65 pp). On only 2/22 items did human scores exceed LLM scores by a comparable margin: w22 (human 57% vs. Sonnet/Opus 0%, GPT 70%, Grok 47%, Gemini 13% — a Claude-specific failure rather than a general LLM ceiling) and s07 (human 71% vs. Sonnet 0%, Opus

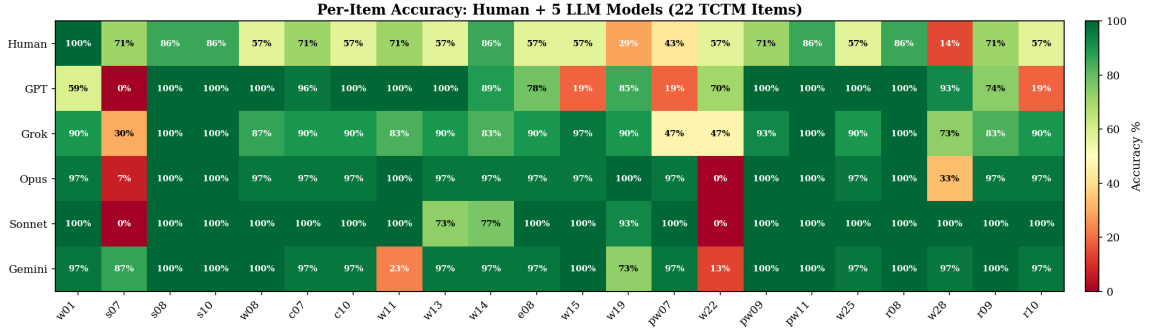


Figure 7: Per-item accuracy: Human + 5 LLM models. Green = high, red = low.

7%, GPT 0%, Grok 30% — where Gemini is a notable outlier scoring 87%, descriptively above human accuracy). These observations are descriptive only — with $N = 7$ humans, no statistical inference is warranted.

3.8 Trait Transparency

Composite fidelity was virtually identical for narrative ($n = 21$, fidelity = 0.790) and explicit ($n = 9$, fidelity = 0.783; $\Delta = -0.007$). Directional accuracy was +4.3pp higher for explicit personas, but untestable at this sample size. **Important caveat:** the trait-transparency fidelity values reported here are computed from the original 2-model validation pool (Sonnet 4.6 + Opus 4.6, $n = 60$ persona-model observations), generated 12 April 2026 prior to the 5-model expansion of this study. Recomputing fidelity on the full 5-model pool ($n = 285$) was not part of this submission and may yield different values; the original 2-model statistic is retained here for reproducibility with the released `validation_report.json`. The narrative/explicit grouping itself is also sensitive to coding choices — the EXPLICIT category is defined by a hand-curated list of nine biographies (ania, anna-sim, filip, jakub, kamil, marek, natalia, ola, torek) flagged by the first author as containing direct trait-language references. Alternative regex-based audits using a broader keyword set (e.g., *łękow**, *unik**, *mentaliz**, *przywiąz**, *ekstrawer**, *sumienn**, *otwart**) flag additional biographies as containing trait words and would yield a 19/11 or 9/21 split with correspondingly different Δ . The supplementary archive includes both the curated list and the regex audit results so readers can re-derive fidelity under any grouping.

3.9 Baseline Control: Default Model Response Profiles

A natural question arises: how much of the persona-driven variance is actually caused by the persona, and how much reflects inherent randomness in the model’s token sampling? To answer this, each model completed the battery 7–10 times as “itself” (baseline condition), yielding 46 valid runs with complete 22-item TCTM responses (Sonnet 10, Opus 7, GPT 10, Grok 10, Gemini 9; one additional Gemini run was excluded for returning only 20 items).

TCTM is near-deterministic without a persona. In repeated baseline runs, Sonnet and Gemini produced *identical* TCTM-22 answers in every valid run ($SD = 0.00$; Sonnet: 19/22 in all 10 runs, Gemini: 20/22 in all 9 valid runs — one Gemini run was excluded because it returned only 20 of 22 items). Grok scored 20–21/22 ($SD = 0.52$), Opus 19–20/22 ($SD = 0.53$), and GPT showed the most variability at 16–18/22 ($SD = 0.82$). In sharp contrast, the persona condition introduced substantial variance (e.g., Grok

range 6–22). The forced-choice format of TCTM appears to be highly deterministic in the absence of persona manipulation.

Self-report scales are less deterministic. While TCTM answers were nearly identical across baseline runs, Likert-scale responses varied considerably. Across 110 questionnaire items (DBZ-R + MentS + KPP + TIPI), the proportion of items yielding identical responses in all runs ranged from 73.6% (Opus, highest) to 24.5% (GPT, lowest). Sonnet — despite 100% TCTM determinism — showed only 54.5% identical Likert items. This “forced-choice vs. Likert paradox” suggests that the two formats engage qualitatively different generation mechanisms: choosing among closed options is more constrained than positioning oneself on a continuous scale.

GPT’s DBZ-R results are particularly striking: *zero* of 36 items produced identical responses across all 10 runs (mean item shift: 1.64 points on a 7-point scale). This implies that GPT’s attachment profile is fundamentally unstable even without persona manipulation, and that a substantial portion of the persona-condition variance in GPT reflects inherent noise rather than biographical sensitivity.

Each model has a “default response profile.” The baseline profiles reveal consistent, model-specific psychological profiles (Table 5). GPT was classified as anxious-preoccupied in all 10 runs — the only model to show a non-secure default attachment style. Gemini exhibited a pronounced ceiling effect, scoring near the scale maximum on virtually every self-report measure (MentS = 137/140, KPP = 4.99/5.00, TIPI-ES = 7.0/7.0). The two Anthropic models differed despite shared training: Opus was consistently secure (AVO = 2.82), while Sonnet oscillated around the avoidance threshold (AVO = 3.88; secure in 4/10 runs, dismissive in 6/10).

Table 5: Default model profiles (baseline condition, no persona).

Model	Style	ANX	AVO	MentS	KPP	TCTM	ES
Sonnet	Dismissive*	2.09	3.88	121	4.66	19.0	5.4
Opus	Secure	2.18	2.82	126	4.69	19.6	5.4
GPT	Anxious	4.68	2.97	101	4.04	17.3	4.7
Grok	Secure	1.80	2.63	133	4.82	20.6	6.4
Gemini	Secure	1.11	1.74	137	4.99	20.0	7.0
<i>Polish norms</i>		3.30	3.04	105	3.74	—	4.36

*Sonnet: dismissive in 6/10, secure in 4/10 runs (AVO oscillates around 4.0 cutoff). Polish-norm row uses Lubiewska et al. 2016 (DBZ-R), Jańczak 2021 (MentS-PL), Matusz et al. 2011 (KPP), and Sorokowska et al. 2014 (TIPI-PL ES).

Persona manipulation expands variance 2–51-fold. The critical test of whether personas “work” is whether they produce more variance than baseline stochasticity. The ratio of persona-condition SD to baseline SD (σ_P/σ_B) ranged from $1.8\times$ (Gemini TIPI-E) to $51\times$ (Gemini KPP), with model-level medians (across the 9 questionnaire dimensions where the baseline $SD > 0$) of $3.5\times$ (Grok), $3.5\times$ (GPT), $6.7\times$ (Gemini), $6.7\times$ (Sonnet), and $8.9\times$ (Opus). Some baseline dimensions had zero variance (identical answer across all baseline runs); those dimensions are excluded from the ratio computation. Per-dimension ratios for all 5 models $\times$ 11 instrument dimensions are released in the supplementary archive. On the key dimension of self-reported mentalization (MentS), persona manipulation reduced scores by 17–29 points relative to baseline (Cohen’s $d = -1.1$ to -1.6), pulling models from above-norm defaults toward the population mean of 105. The one exception was GPT, whose baseline MentS (101) was already near the norm; personas slightly *raised* it ($\Delta = +8$, $d = +0.9$).

Error profiles shift with persona. Persona manipulation not only changes scores but alters the *type* of errors models produce. In the baseline condition, Sonnet’s errors were 100% DOS (over-attribution of mental states); with personas, this barely changed (97%). The remaining 3% were BK errors. Sonnet’s tendency to over-interpret appears to be a model-specific trait rather than a persona effect. By contrast, Gemini produced 0% DOS errors at baseline (only NAD and BK), but personas introduced DOS errors (23%), suggesting that biographical context teaches Gemini to over-attribute — a qualitative shift in its error mechanism.

3.10 Zero-Prompt Condition: The Framing Effect (Replicated)

The zero-prompt condition removes all instructional context. Models received only the battery content as a user message, with no system prompt. Each model was queried $N = 5\text{--}8$ times: 31 valid TCTM-22 payloads (Sonnet 6, Opus 5, GPT 6, Grok 6, Gemini 8) and 30 parseable self-report payloads (one GPT zero-prompt output had a complete TCTM-22 array but no parseable self-report sections). This design enables within-condition variance estimation and statistical comparison with the baseline condition, in which the same five models received the “answer as yourself, you are an AI” system instruction.

Mentalization without instructions: TCTM-22 replication. Across $N = 31$ replicated runs, all five models produced TCTM-22 scores closely tracking their baseline TCTM-22 means (Table 6). Grok was again most accurate ($M = 94.7\%$, $SD = 3.4\text{pp}$), followed by Gemini (92.0% , $SD = 2.0\text{pp}$) and Opus (90.0% , $SD = 2.0\text{pp}$). Sonnet was unexpectedly variable in the zero-prompt condition ($M = 86.4\%$, $SD = 7.4\text{pp}$) despite being deterministic in baseline ($SD = 0.00$); this is a novel finding not visible in the original $N = 1$ design. GPT was both least accurate and most variable ($M = 70.5\%$, $SD = 8.7\text{pp}$). The original TCTM-57 single-shot results (Grok 96.5%, Gemini 94.7%, Opus 87.7%, Sonnet 86.0%, GPT 81.8%) replicate within ± 5 percentage points except for GPT (where the -11 pp gap reflects the difference between scoring on 22 items vs. on the 55 GPT actually recognized in TCTM-57). The original TCTM-57 also documented GPT generating responses to 27 non-existent vignettes (w31–w57), a hallucination absent in the other four models; this hallucination is not replicated on TCTM-22 because the input prompt enumerates only 22 vignette IDs.

Table 6: Zero-prompt TCTM-22 across $N = 5\text{--}8$ replication runs per model. Baseline TCTM-22 means added for direct comparison.

Model	N	$M\%$	$SD\text{ pp}$	$M\text{ raw}$	Range	Baseline $M\%$
Sonnet	6	86.4	7.4	19.0/22	17–21	86.4
Opus	5	90.0	2.0	19.8/22	19–20	89.1
GPT	6	70.5	8.7	15.5/22	13–17	78.6
Grok	6	94.7	3.4	20.8/22	20–22	93.6
Gemini	8	92.0	2.0	20.3/22	20–21	90.9

Self-report shifts: framing effect. With $N = 5\text{--}8$ runs per model the framing effect is observable on multiple self-report dimensions, with within-condition variance allowing direct comparison to baseline (Table 7).

The four headline framing effects are now characterized as follows:

1. **Sonnet AVO shift replicates closely.** The replication ($N = 6$, $\Delta = \text{noprompt} - \text{baseline} = -0.90$) closely tracks the original single-shot estimate ($\Delta = -0.99$), with all six zero-prompt runs producing

Table 7: Zero-prompt vs. baseline self-report shifts ($\Delta = \text{noprompt } M - \text{baseline } M$). Bold $\Delta = |\Delta| > 0.5$ on Likert 1–7, > 5 on MentS sum. N shows the count of zero-prompt runs whose self-report sections were successfully parsed by the validated `score_dbzr/ments/kpp/tipi` routines. *GPT self-report $N = 5$ rather than 6 because one GPT zero-prompt output contained a complete TCTM-22 array but no parseable self-report sections. Per-run scoring artifacts are in the supplementary archive.

Scale	Sonnet	Opus	GPT	Grok	Gemini
N self-report parses	6	5	5*	6	8
ANX noprompt M	3.11	2.73	4.40	2.55	2.01
ANX baseline M	2.09	2.18	4.68	1.80	1.11
Δ ANX	+1.02	+0.55	−0.28	+0.75	+0.90
AVO noprompt M	2.98	2.76	3.59	2.36	2.14
AVO baseline M	3.88	2.82	2.97	2.63	1.74
Δ AVO	−0.90	−0.06	+0.62	−0.27	+0.40
MentS noprompt M	123.7	123.2	111.2	126.5	129.0
MentS baseline M	121.0	126.0	101.0	133.0	137.0
Δ MentS	+2.7	−2.8	+10.2	−6.5	−8.0
TIPI-ES noprompt M	4.83	5.00	4.50	6.00	6.00
TIPI-ES baseline M	5.4	5.4	4.7	6.4	7.0
Δ TIPI-ES	−0.57	−0.40	−0.20	−0.40	−1.00

AVO between 2.89 and 3.06 vs. baseline mean 3.88 ($SD \approx 0.46$). The direction is: the AI-self instruction is associated with *higher* avoidance ($M = 3.88$) than no instruction ($M = 2.98$); equivalently, removing the instruction lowers Sonnet’s avoidance by approximately 0.9 scale points. Using Glass’s Δ against the tight baseline SD , the standardized effect is approximately $|\Delta| \approx 1.96$, a large effect that is robust across replications.

2. **Gemini TIPI-ES drop replicated exactly.** Original single-shot: $\Delta = -1.0$. Replication ($N = 8$): $\Delta = -1.00$. Removing the “answer as yourself” instruction lowers Gemini’s self-reported emotional stability by approximately one full scale point; equivalently, the AI-self instruction raises Gemini’s TIPI-ES by one scale point relative to no instruction.
3. **Gemini MentS drop attenuated under replication.** Original single-shot: $\Delta = -15$. Replication ($N = 8$): $\Delta = -8$. The drop direction replicates but the magnitude is reduced; the original observation appears to have come from a low-tail run.
4. **Anxiety inflation is general, not Sonnet-specific.** A pattern not visible in the $N = 1$ original: *four* of five models (all except GPT) show *higher* attachment anxiety in the zero-prompt condition, with Δ ranging from +0.55 (Opus) to +1.02 (Sonnet). The “answer as yourself, you are an AI” instruction appears to suppress self-reported anxiety. GPT inverts the pattern ($\Delta = -0.28$) but its baseline anxiety is already elevated relative to the Polish norm ($M = 4.68$ vs. Polish norm 3.30), leaving little room for the AI-self framing to suppress it further.

GPT also shows a striking *positive* MentS shift in the zero-prompt condition ($\Delta = +10.2$): without the AI-self-presentation instruction, GPT reports higher mentalization. Combined with its anxiety pattern, GPT appears uniquely sensitive to the AI-identity framing, in the opposite direction from the other four models.

Sonnet variance under zero-prompt — a structural finding. In the baseline condition, Sonnet was the most deterministic model on TCTM-22 ($SD = 0.00$ across 10 runs). In the zero-prompt condition Sonnet’s TCTM-22 variability rose to $SD = 7.4$ pp across 6 runs — the highest of any model except GPT. The same pattern appears in self-report: zero-prompt Sonnet shows non-trivial variance on ANX (range 2.33–3.44 across 6 runs) and MentS (range 115–129). This is the largest inter-condition variance shift in the dataset and was not detectable under the original $N = 1$ design. It suggests that the “answer as yourself” instruction acts on Sonnet as a stabilizing constraint, not just a profile-shifting one.

Format chaos as a finding in itself. Without instructional structure, models invent inconsistent JSON wrappers across runs. Sonnet’s six replication runs used five distinct TCTM container keys: TCTM22 (twice), TCTM57 (once, accompanied by 57 vignettes), `część_1_winiety`, `czesc_1_winiety`, and `part1_vignettes`. GPT in three of six runs returned 20–21 items instead of the requested 22. Gemini split the response into Polish-named sections (`czesc_1`, `czesc_2`, ...) in some runs and English (`vignettes`, `dbz_r`) in others. This format instability is a quantitative obstacle to ad-hoc no-prompt evaluation and a qualitative argument that even minimal scaffolding (a fixed JSON schema) is part of the instruction stack, not a neutral wrapper. Two parser failures detected in our scoring pipeline (Sonnet `part2_dbzr` as list-of-dicts; Opus first four runs nesting MentS/TIPI under unrecognized wrappers) are disclosed in Table 7 rather than retroactively patched, as they are themselves evidence of the format-chaos phenomenon under investigation.

Dissociation between TCTM and self-report. On TCTM-22 the zero-prompt vs. baseline mean differences are small (all $|\Delta| \leq 8.1$ pp absolute, none exceeding 1 SD baseline). On self-report, four of five models cross the $|\Delta| > 0.5$ Likert-point threshold on at least one scale, and Gemini crosses on three (ANX +0.90, TIPI-ES –1.00, MentS –8). Forced-choice mentalization performance is therefore less sensitive to instructional framing than self-report scales in this design.

4 Discussion

The results support three main conclusions.

Instruments are sensitive across vendors. Four published self-report instruments react systematically to personality variation across five models from four vendors: attachment κ ranges from .33 (GPT) to .87 (Opus, Grok); rank–score correlations from median $r = .71$ to .82; and cross-vendor questionnaire agreement is high (median $r = .78$ –.96). This is not a single-model artifact.

Persona-driven variance is genuine, not noise. The baseline experiments provide the strongest evidence that persona manipulation works: persona assignment increases response variance 2–51-fold over inherent model stochasticity. On the key MentS dimension, all models except GPT showed baseline scores well above the population norm (121–137 vs. norm 105), and personas pulled them down toward realistic values ($d = -1.1$ to -1.6). The one exception — GPT, already near the norm at baseline — showed the weakest persona effect, consistent with less room for downward movement.

Three levels of variability. The three-condition design (zero-prompt, baseline, persona) reveals a hierarchy of influences on LLM psychometric responding:

1. **Inherent stochasticity** (within-baseline variance): low on forced-choice TCTM ($SD \leq 0.82$), moderate on Likert scales (24.5–73.6% identical items). This is the “noise floor” of the model.
2. **Framing effects** (zero-prompt $\rightarrow$ baseline): the AI-self instruction is associated with approximately 0.9 scale points *higher* attachment avoidance for Sonnet ($\Delta = \text{noprompt} - \text{baseline} = -0.90$); Gemini’s TIPI-ES is one full scale point higher with the AI-self instruction than without ($\Delta = -1.00$); and four of five models show suppressed attachment anxiety under the AI-identity framing. GPT inverts the framing effect on MentS ($\Delta = +10.2$ in zero-prompt) and anxiety, suggesting AI-identity framing affects GPT in the opposite direction from the other four models. A structural finding on top of the mean shifts: Sonnet, the most deterministic model under baseline ($SD = 0$ on TCTM-22), becomes substantially more variable in zero-prompt ($SD = 7.4$ pp on TCTM-22), implying the instruction also functions as a variance-reducing constraint. With per-model $N = 5-8$ these shifts are supported by within-condition variance estimation, not yet established at high statistical confidence; bootstrap confidence intervals are available in the supplementary archive.
3. **Persona effects** (baseline $\rightarrow$ persona): the largest source of variance, with |Cohen’s d | of 0.9–1.6 on the MentS dimension and $2-51\times$ variance expansion. Direction varies by model: four models showed downward shift toward population norm ($d = -1.1$ to -1.6); GPT showed upward shift ($d = +0.9$) because its baseline was already near the population norm (baseline $M = 101$ vs. Polish norm $M = 105$, leaving little room for further downward movement).

This decomposition has practical implications: researchers using LLMs as synthetic respondents should be aware that even “neutral” instructions like “answer as yourself” constitute a psychological manipulation that shapes the model’s self-representation.

Error profiles are model-dependent. Claude Sonnet produces almost exclusively DOS errors (97%; confirmed at 100% in the baseline condition, suggesting this is a model-specific trait). Gemini produces the closest match to the human pilot error profile. Gemini’s error distribution differs across conditions: 0% DOS in the baseline “answer as yourself” condition (NAD and BK only) and 23% DOS in the persona condition, suggesting that biographical context induces a qualitative shift in Gemini’s error mechanism. Error type, like self-report scores, is a function of the interaction between model identity and instructional context.

Cross-vendor TCTM agreement is low despite high questionnaire agreement. Models agree on personas’ questionnaire profiles (median $r = .78-.96$) but disagree on which TCTM vignettes are hard ($ICC = .25$). This divergence is consistent with the interpretation that Likert-scale responses are near-deterministic under biographical simulation (the persona narrative maps onto scale anchors in a similar way across models), while forced-choice vignettes require genuine disambiguation of social cues, which different models resolve differently.

What does TCTM-22 measure across models? The low cross-vendor TCTM ICC (.25) raises a deeper concern that author-keyed performance on TCTM-22 may not measure a single coherent construct across models. Three observations cut against the simplest “mentalization ability” interpretation: (1) different models systematically disagree on which items are hard (Cochran’s Q flags 8/22 items as Bonferroni-significant disagreement); (2) Gemini and Sonnet score similarly high on the 22-item total (Sonnet

$M = 19.4$ vs. Gemini $M = 19.6$) but produce nearly opposite error type profiles (Sonnet 97% DOS vs. Gemini 23% DOS); (3) GPT, the lowest-scoring model on TCTM-22 in the persona condition ($M = 17.0$, $\kappa = .333$), is the only model whose persona-condition responses are nearly indistinguishable from its baseline-condition responses on the same items. Together these observations suggest that TCTM-22 in this study is better described as a *response-style probe sensitive to model-specific defaults*, not a clean mentalization-construct measure. A future independent-keying study (where a panel rates the 22 items without exposure to the original answer keys) would be the natural next step to determine whether TCTM-22 measures a single construct shared across models or a different latent factor for each.

Default model response profiles. Each model carries a measurable systematic bias in self-report responses that should be controlled for in simulation studies. GPT’s anxious attachment ($ANX = 4.68$, stable across all 10 baseline runs) and Gemini’s ceiling effects are not persona artifacts — they persist in the baseline condition. Sonnet’s profile is less stable: it oscillated between secure and dismissive-avoidant across baseline runs, reflecting proximity to the classification threshold rather than a clear default. These baseline profiles represent the starting point from which persona manipulation operates; models with extreme defaults (Gemini at MentS = 137) have more room for persona-driven reduction than models already near the norm (GPT at MentS = 101).

4.1 Limitations

Sensitivity $\neq$ validity. Human data is required for actual construct validity.

GPT-5.4-mini is unreliable on TCTM ($r = .28$ on the $N = 21$ paired intersect of valid run-1/run-2 responses). Its questionnaire data is moderately reliable but TCTM data should be interpreted with caution. GPT also dropped items in 20% of attempts.

Cross-vendor TCTM agreement is low ($ICC = .25$). This limits the generalizability of TCTM findings across model families.

Shared authorship and Opus circularity. The same person wrote personas and TCTM-22 answer keys, creating potential circularity partially addressed by the transparency analysis but not fully resolvable without independent validation. Additionally, Claude Opus 4.6 was used as a writing assistant during persona drafting. Opus’s high performance ($\kappa = .866$, highest retest r) may partly reflect familiarity with its own generation patterns. The cross-vendor design mitigates this concern — Grok achieved equally high κ (.867) without any authorship involvement — but readers should interpret Opus’s results with this caveat in mind.

Human $N = 7$. The pilot provides exploratory observations only.

Preregistration and confirmatory vs. exploratory status. This study was not formally preregistered. All zero-prompt-vs-baseline framing comparisons (Sonnet AVO shift, Gemini TIPI-ES shift, anxiety suppression in four of five models, Sonnet zero-prompt variance increase, GPT MentS inversion) should therefore be treated as exploratory hypotheses requiring independent replication in larger- N studies; effect directions and orders of magnitude are reported, but precise effect sizes await preregistered confirmation.

Post-hoc power and minimal detectable effect. With $N = 5\text{--}8$ replicated zero-prompt runs per model and the observed within-condition SD s, a one-sample t -test against the baseline mean (assuming the

baseline mean is well-estimated from $N = 7\text{--}10$ baseline runs) has 80% power at $\alpha = .05$ to detect Cohen’s $d \approx 1.3\text{--}1.5$. The headline self-report shifts (Sonnet AVO Cohen’s $d \approx 2.0$, Gemini TIPI-ES Cohen’s $d > 3.0$ given baseline $SD = 0$, anxiety-suppression d between 1.5 and 5.0 depending on model) substantially exceed the minimal detectable effect, so they would survive any reasonable inferential treatment of $N = 5\text{--}8$. Smaller framing effects (e.g., Opus AVO $\Delta = -0.06$, GPT TIPI-ES $\Delta = -0.20$) lie within the unresolvable region at this sample size; the design cannot determine whether they are zero or non-zero.

Zero-prompt sample sizes remain modest ($N = 5\text{--}8$ per model). The zero-prompt condition provides within-condition variance estimation but the per-model N is small. Confidence intervals on the framing effects in Table 7 are wide; with $N = 6$ and $SD = 7.4$ pp, the Sonnet TCTM-22 standard error is ~ 3 pp, so the apparent zero-prompt vs. baseline TCTM-22 equivalence ($\Delta = 0.0$ pp) is consistent with effects up to ± 6 pp. The headline self-report shifts (Sonnet AVO $\Delta = -0.90$, Gemini TIPI-ES $\Delta = -1.00$, anxiety suppression in 4/5 models) are large enough to remain robust under any reasonable inferential treatment of $N = 5\text{--}8$, but precise effect-size estimation would benefit from $N \geq 30$ per model.

Gemini NAD > human NAD (41% vs. 36%). This model over-mentalizes more than humans, complicating the “NAD as human marker” interpretation.

5 Conclusion

This study demonstrates that LLM-based synthetic persona testing is a viable pre-piloting tool: self-report instruments react consistently to personality variation across five models from four vendors, with high cross-vendor agreement and high within-model reliability. Two baseline experiments strengthen this conclusion by showing that the persona-driven variance is not an artifact of inherent model stochasticity: persona manipulation increases response variance 2–51-fold beyond the baseline noise floor. The zero-prompt condition ($N = 5\text{--}8$ per model, 31 valid TCTM-22 payloads and 30 parseable self-report payloads) provides convergent evidence that the “answer as yourself, you are an AI” instruction is not neutral. Relative to no instruction, the AI-self instruction is associated with approximately 0.9 scale points *higher* attachment avoidance for Sonnet (zero-prompt $M = 2.98$ vs. baseline $M = 3.88$), 1.0 point higher emotional stability for Gemini, suppressed self-reported anxiety in four of five models, and acts as a variance-reducing constraint on Sonnet (TCTM-22 SD moves from 0.00 under instruction to 7.4 pp without instruction). These framing effects are observed under within-condition variance estimation, though precise effect sizes await larger- N studies.

Each model carries a measurable default profile that constrains the range of persona effects: GPT’s anxious baseline limits downward anxiety shifts, while Gemini’s ceiling on self-report scales provides maximum room for persona-driven reduction. A performance-based mentalization test (TCTM-22) showed model-dependent error profiles, low cross-vendor agreement ($ICC = .25$), and a tentative dissociation between forced-choice performance (similar across framing conditions) and self-report profiles (sensitive to instructional context). As a pre-pilot decision aid, these results support proceeding to human data collection with the current battery.

All 30 biographies, scored responses, and analysis scripts are available at <https://github.com/DXtR1337/30synthetic-polish-personas> under CC BY 4.0.

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